

Inkguard-AI

AI-powered content creation platform with built-in safety — writing tools protected by real-time plagiarism, brand voice, and prompt-injection detection, powered by IBM Granite via watsonx.ai.

InkGuard AI — IBM

1. Problem Brief

What real business problem does this solve?

Students, content creators, and small businesses need to produce written content quickly (essays, blogs, emails, CVs, marketing copy) but face two growing risks: (1) AI-generated or plagiarized content getting flagged or penalized by schools/platforms, and (2) no easy way to check content originality, brand consistency, or safety before publishing. Existing tools solve either "creation" or "detection" separately, forcing users to juggle multiple disconnected apps.

Who is the target user/persona?

Students submitting essays/assignments who need to verify originality before submission
Content creators and small business owners producing blogs, emails, and marketing material
Educators who want a lightweight way to check submitted work for AI-generated content

What is the current pain point and how does this improve it?

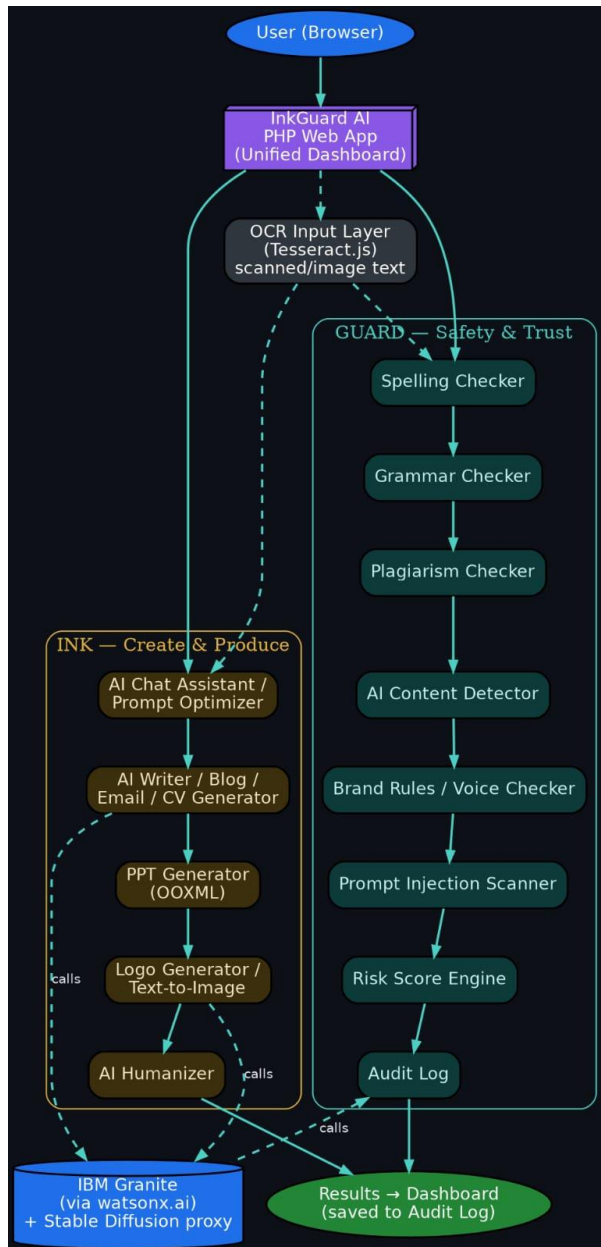
Currently, users must use separate tools — one for grammar/plagiarism checking, another for AI detection, another for content generation, another for design (CVs, PPTs, logos). This is slow, inconsistent, and provides no unified trust layer.

InkGuard AI combines 20 tools across two integrated pillars — GUARD (safety & trust: spelling, grammar, plagiarism, AI-content detection, risk scoring, audit logs, prompt-injection scanning) and INK (create & produce: AI writer, blog/email/CV generation, PPT and logo generation, text-to-image, humanizer) — into a single dashboard, so content is created and verified in one flow instead of many.

Industry/domain context

EdTech and content-creation tooling, with a focus on academic integrity and responsible AI-assisted writing — relevant given the rapid rise of AI detection requirements in schools and publishing platforms.

2. Architecture Diagram (description) System design showing data flow:



Which CLI/AI skills are used and how they connect:

Not built on Cortex Code CLI — this is a standalone PHP web application. IBM Granite (via watsonx.ai) is called directly through PHP backend proxies for content generation and detection; Stable Diffusion/Hugging Face is used for image generation.

Data sources (structured/unstructured):

- Unstructured: free text input, uploaded documents/images (OCR)
- Structured: risk scores, audit log entries, user-submitted forms (CV data, brand rules)

How modular components plug together:

Each of the 20 tools is a self-contained module sharing a common PHP backend and authentication layer, so GUARD and INK tools can be combined in one workflow (e.g., generate a blog post with INK, then immediately run it through GUARD's plagiarism and AI-detector checks before publishing).

3. Impact Statement Measurable outcomes:

- Reduces the number of separate tools a student/creator needs from 5+ down to 1 dashboard
- Cuts content-review time by combining generation and verification into a single pass instead of sequential tool-switching
- Provides a quantifiable Risk Score so users get an objective originality/safety measure instead of guesswork

Scalability potential:

The modular architecture (20 independent tools sharing one backend) allows new checks or generators to be added without reworking the core system. It can scale from individual students to institutional use (schools/universities running bulk originality checks) or business teams needing brand-consistent content at volume.

How this extends beyond the demo:

InkGuard AI is already live at ink.fredev.app and free for students to try. Beyond the hackathon, it can integrate with LMS platforms (Google Classroom, Moodle) for automatic assignment screening or offer an API for other apps to plug in GUARD's trust-layer checks directly.